Test Series Code

03

All India Test Series
NEET (2025-26)

SET

01

DURATION: 180 Minutes | **DROPPER** | **M. MARKS: 720**

DATE: 07/12/2025

SYLLABUS

Physics:	Rotational Motion, Gravitation, Mechanical Properties of Solids
Chemistry:	Thermodynamics & Thermochemistry, Organic Chemistry: IUPAC Nomenclature
Botany:	The Living World, Biological Classification
Zoology:	Excretory Products & their Elimination, Locomotion & Movement

GENERAL INSTRUCTIONS:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours** duration and the Test Booklet contains **180** multiple choice questions (four options with a single correct answer).
3. There is only one correct response for each question.
4. Each **correct** answer will give **4 Mark** while **1 Mark** will be **deducted** for a **wrong** answer. The maximum marks are 720.
5. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

OMR INSTRUCTIONS:

1. Use only blue/black dark ballpoint pen.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

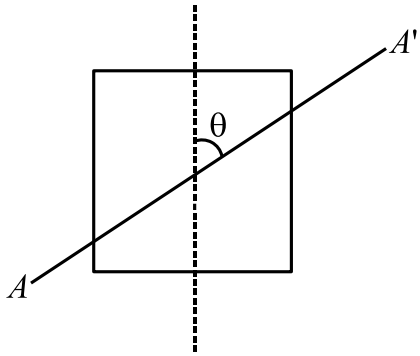
Name of the Student (In CAPITALS): _____

Roll Number: _____

OMR Bar Code Number: _____

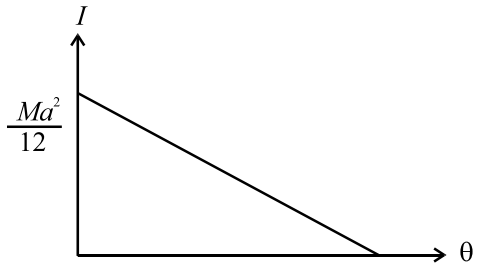
Student's Signature: _____ **Invigilator's Signature:** _____

1. A uniform square sheet of mass '*M*' and side '*a*' rotates about an axis *AA'* that lies in the plane of the sheet and passes through its center of mass. The axis makes an angle θ with a vertical line of symmetry as shown.

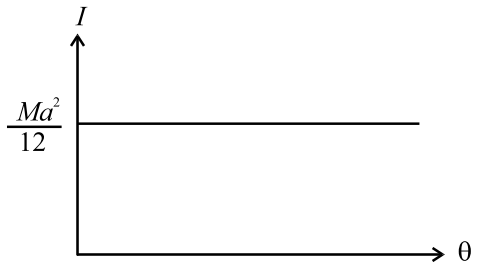


If the angle θ is continuously increased, which graph best represents the relationship between the moment of inertia (*I*) about this axis and the angle θ?

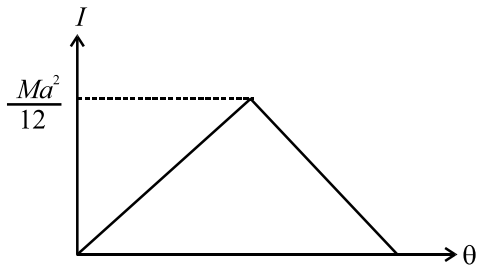
(1)



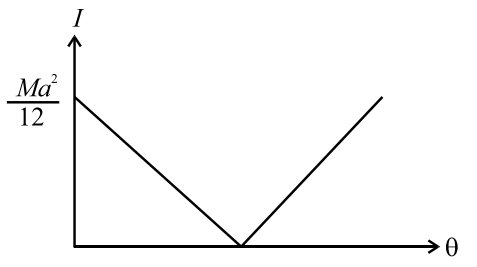
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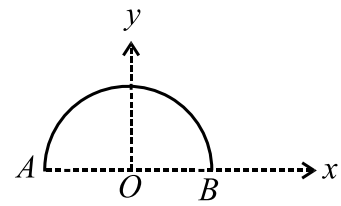
(3)



(4)



2. A thin wire of length 1 m is bent to form a semicircle as shown in the figure. The magnitude of the gravitational field at the center of the semicircle (point *O*) is 4πG N/kg, where *G* is the universal gravitational constant. What is the mass of the wire?



(1) 1 kg (2) 2 kg

(3) 4 kg (4) π kg

3. During a tensile strength test of a steel cable, a student increases the load such that the stress on the cable doubles. Assuming the test is conducted within the material's elastic limit, what is the effect on the Young's Modulus of the steel?

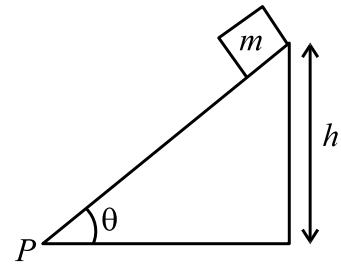
(1) It doubles

(2) It remains constant

(3) It is halved

(4) It quadruples

4. A small object of mass m is placed at the top of a fixed inclined plane of inclination θ. The object is at a vertical height *h* above the ground as shown. What is the magnitude of the torque produced by the object's weight about the vertex *P*? (neglect object size) (*g* is acceleration due to gravity)



(1) *mgh*

(2) *mgh* sin(θ)

(3) *mgh* tan(θ)

(4) *mgh* cot(θ)

5. Three point masses are placed in the *x*-*y* plane. A mass $m_1 = 2$ kg is at the origin $(0,0)$ m, a mass $m_2 = 9$ kg is at $(3,0)$ m, and a mass $m_3 = 32$ kg is at $(0,4)$ m. Masses m_2 and m_3 are fixed in place, while mass m_1 is free to move. What is the initial acceleration of mass m_1 due to the gravitational attraction from the other two masses? (Assume G is the universal gravitational constant in SI units).

(1) $G(\hat{i} + 2\hat{j})$ m/s^2

(2) $G(2\hat{i} + 4\hat{j})$ m/s^2

(3) $G(\hat{i} - 2\hat{j})$ m/s^2

(4) $2G(\hat{i} + \hat{j})$ m/s^2

6. A solid sphere has mass *M* and radius *R*. A small test mass *m* is placed at a distance 3*R* from the center. What is the magnitude of the gravitational force exerted by the sphere on the test mass *m*?

(1) $\frac{GMm}{R^2}$ (2) $\frac{GMm}{2R^2}$

(3) $\frac{GMm}{9R^2}$ (4) Zero

7. A mechanical arm is acted upon by a force vector $F = (\hat{i} + 2\hat{j} - 3\hat{k})$N at a point with position vector $r_1 = (3\hat{i} + \hat{j} + 2\hat{k})$m . What is the torque produced by this force about a pivot point located at $r_2 = (2\hat{i} - \hat{j} + \hat{k})$m ?

(1) $(8\hat{i} - 4\hat{j})$Nm

(2) $(-8\hat{i} + 4\hat{j})$Nm

(3) $(-4\hat{i} - 4\hat{j} + 2\hat{k})$Nm

(4) $(-8\hat{i} - 4\hat{j})$Nm

8. Young's modulus (in N/m^2) for a perfectly rigid body is:

(1) Zero (2) Infinite

(3) 0.5 (4) 1

9. A newly discovered planet, "Kepler-*X*," has a mass M_p and a radius R_p. An object of mass *m* is dropped near its surface. The acceleration due to gravity it experiences is g_p. If the planet's mass were doubled ($2M_p$) and its radius were halved ($R_p/2$), what would be the new acceleration due to gravity, g_{new}?

(1) 2 g_p (2) 4 g_p

(3) 8 g_p (4) $\frac{g_p}{2}$

10. Two wires, *P* and *Q*, are made of the same alloy and have the same volume. Wire *P* has a cross-sectional area *A* and wire *Q* has a cross-sectional area 4*A*. If a force *F* is required to stretch wire *Q* by an amount Δl, what force is needed to stretch wire *P* by the same amount Δl?

(1) 16*F*

(2) 4*F*

(3) $\frac{F}{4}$

(4) $\frac{F}{16}$

11. A solid circular disc of mass 50 kg and radius 1 m is rotating about its central axis perpendicular to the plane of the disc with angular speed 4 rad/s. The absolute value of external work done required on the disc to stop it is:

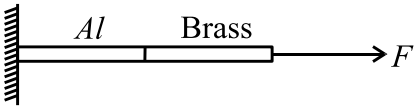
(1) 200 J

(2) 400 J

(3) 600 J

(4) 800 J

12. An aluminum wire and a brass wire of the same length are connected end-to-end. This composite wire is stretched by a force F as shown. If both wires have the same cross-sectional area, what can be said about the stress and strain in each wire? (Young's Modulus for brass is greater than for aluminum: $Y_{brass} > Y_{alum}$).



(1) Same stress, same strain.

(2) Same stress, different strain.

(3) Different stress, same strain.

(4) Different stress, different strain.

13. A body moves in a circle with angular acceleration $\alpha = 6t^2 - 2t$ (where *t* is in second and α is in rad/s^2). At *t* = 0, the ball has angular velocity of 10 rad/s. What is the angular velocity of body at *t* = 5 s?

(1) 235 rad/s

(2) 225 rad/s

(3) 125 rad/s

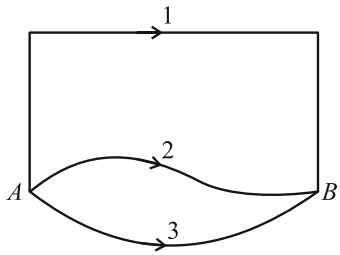
(4) 50 rad/s

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14. Elastic forces:
(1) Are always conservative
(2) Are not always conservative
(3) Are never conservative
(4) None of these

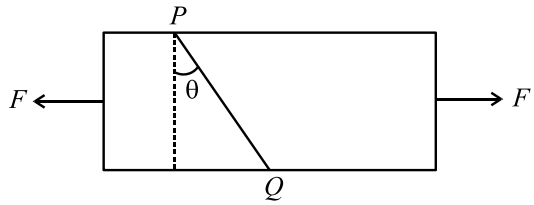
15. An object of mass m is lifted from the surface of the Earth (radius R) to an altitude of $h = \frac{R}{2}$. What is the change in its gravitational potential energy? (g is acceleration due to gravity)
(1) $\frac{1}{3}mgR$ (2) $\frac{1}{2}mgR$
(3) mgR (4) $\frac{2}{3}mgR$

16. A gravitational field is present in a region where a mass is shifted from A to B through different paths as shown. If W_1, W_2 and W_3 represent the work done by the gravitational force along the respective paths, then:



(1) $W_1 < W_2 < W_3$
(2) $W_1 = W_2 = W_3$
(3) $W_1 > W_2 > W_3$
(4) $W_1 > W_3 > W_2$

17. A rectangular block is subjected to a uniaxial tensile force F along its axis. An engineer needs to determine the plane PQ where the shear stress is at its absolute maximum. At what angle θ shown in figure, shear stress is maximum?



(1) $\frac{\pi}{4}$ (2) $\frac{\pi}{3}$
(3) $\frac{\pi}{6}$ (4) $\frac{\pi}{2}$

18. The axis X and Z in the plane of a planar body are mutually perpendicular and Y-axis is perpendicular to its plane. If the moment of inertia of the body about X and Y axes is respectively 30 kg m^2 and 40 kg m^2 then $M.I.$ about Z-axis in kg m^2 will be:
(1) 70 (2) 50
(3) 10 (4) Zero

19. A circular ring of wire of mass M and radius R is making n revolutions/sec about an axis passing through a point on its rim and perpendicular to its plane. The kinetic energy of rotation of the ring is given by:
(1) $4\pi^2MR^2n^2$ (2) $2\pi^2MR^2n^2$
(3) $\pi^2MR^2n^2$ (4) $8\pi^2MR^2n^2$

20. A particle of mass 30 g experiences a gravitational force of 6.0 N along positive x-direction. The gravitational field at that point is:
(1) $-200\ \hat{j}$ N/kg
(2) $200\ \hat{j}$ N/kg
(3) $-200\hat{i}$ N/kg
(4) $200\hat{i}$ N/kg

21. A thin, uniform spherical shell has a radius of 7 cm. The magnitude of the gravitational field at a distance of 10 cm from its center is 8 N/kg. What is the magnitude of the gravitational field at a distance of 15 cm from its center?
(1) 18 N/kg
(2) 12 N/kg
(3) $\frac{32}{9}$ N/kg
(4) $\frac{16}{3}$ N/kg

22. In an arrangement four particles, each of mass 2 kg are situated at the coordinate points (3, 2, 0) m, (1, –1, 0) m, (0, 0, 0) m and (–1, 1, 0) m. The moment of inertia (in kg-m^2) of this arrangement about the Z-axis will be:
(1) 8
(2) 16
(3) 43
(4) 34

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23. A rotating table completes one rotation in 10 sec and its moment of inertia is 100 kg-m^2. A person of 50 kg mass stands at the centre of the rotating table. If the person moves 2 m from the centre, the angular velocity of the rotating table in rad/sec. will become:

(1) $\frac{2\pi}{30}$

(2) $\frac{20\pi}{30}$

(3) $\frac{4\pi}{3}$

(4) 2π

24. Copper of fixed volume V is drawn into wire of length l. When this wire is subjected to a constant force F, the extension produced in the wire is Δl. Which of the following graphs is a straight line?

(1) Δl versus $\frac{1}{l}$

(2) Δl versus l^2

(3) Δl versus $\frac{1}{l^2}$

(4) Δl versus l

25. The rotational kinetic energy of a rigid body of moment of inertia 5 kg-m^2 is 10 joules. The angular momentum about the axis of rotation would be:

(1) 100 joule-sec

(2) 50 joule-sec

(3) 10 joule-sec

(4) 2 joule -sec

26. Two point objects of mass $2x$ and $3x$ are separated by a distance r. Keeping the distance fixed, how much mass should be transferred from $3x$ to $2x$, so that gravitational force between them becomes maximum?

(1) $\frac{x}{4}$

(2) $\frac{x}{3}$

(3) $\frac{x}{2}$

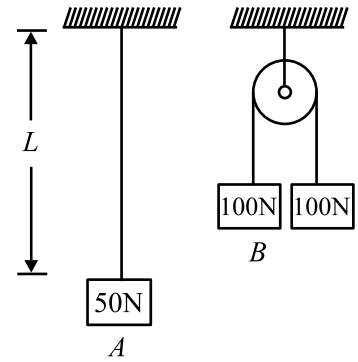
(4) $\frac{2x}{3}$

27. A block of mass 'm' hangs from a uniform wire of mass 'M', area of cross-section A length L and Young's modulus Y. The total elongation in the wire is: (g is acceleration due to gravity)

(1) $\frac{mgL}{AY}$ (2) $\frac{mgL}{2AY}$

(3) $\frac{mgL}{AY}+\frac{MgL}{2AY}$ (4) $\frac{mgL}{2AY}+\frac{MgL}{AY}$

28. In an experiment (Case A), a steel wire of length L is suspended from the ceiling. When a block weighing 50 N is attached to its free end, the wire is observed to elongate by 2.0 mm. In a second experiment (Case B), the same wire of the same length is passed over a frictionless pulley, and blocks of 100 N are suspended on both sides, as shown. What will be the elongation of the wire in Case B?



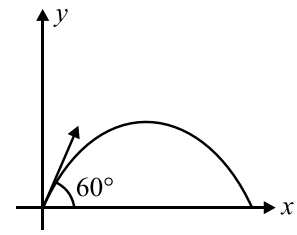
(1) 1.0 mm

(2) 2.0 mm

(3) 4.0 mm

(4) 8.0 mm

29. A particle of mass 2 kg is projected at an angle of 60° above the horizontal, with a speed 20 m/s. The magnitude of angular momentum of the particle about point of projection, when it is at maximum height will be: (g = 10 m/s^2)



(1) 300 kg m^2/s

(2) $120\sqrt{3}$ kg m^2/s

(3) $40\sqrt{3}$ kg m^2/s

(4) 150 kg m^2/s

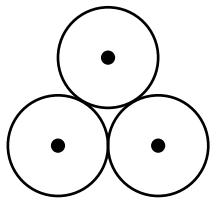
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30. Two point masses, one of mass m and the other of mass $2m$, are fixed at a separation distance of r. Consider an axis of rotation that is perpendicular to the line segment connecting the two masses. What is the minimum possible moment of inertia this system can have with respect to such an axis?

(1) $\left(\frac{1}{2}\right)mr^2$ (2) $\left(\frac{2}{3}\right)mr^2$

(3) $\left(\frac{3}{2}\right)mr^2$ (4) $2\,mr^2$

31. Three solid spheres of mass M and radius R are placed in contact as shown in figure. Find the gravitational potential energy due to interaction of masses in the system?



(1) $-\frac{3}{2}\frac{GM^2}{R}$ (2) $\frac{3}{2}\frac{GM^2}{R}$

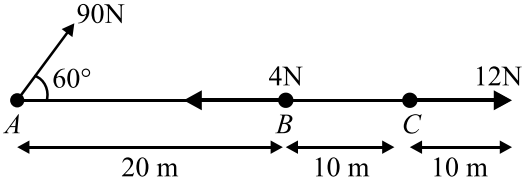
(3) $\frac{5}{2}\frac{GM^2}{R}$ (4) $-\frac{5}{2}\frac{GM^2}{R}$

32. A ladder rests against a frictionless vertical wall, with its upper end 6 m above the ground and the lower end 4 m away from the wall. The weight of the ladder is 500 N and its *C.G.* at 1/3rd distance from the lower end along the ladder's length. Wall's reaction will be closest to, (in Newton)

(1) 111 (2) 333

(3) 222 (4) 129

33. Net torque about point A by all the forces in the given arrangement is:



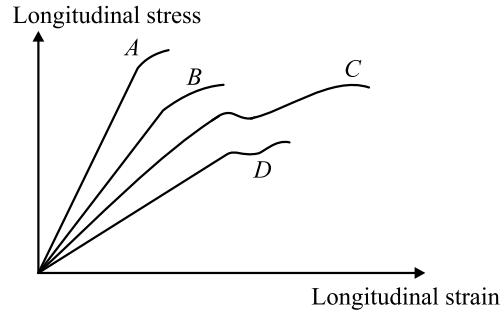
(1) 120 N-m

(2) 80 N-m

(3) 90 N-m

(4) 0 N-m

34. Longitudinal stress v/s longitudinal strain curve for four different materials A, B, C and D is shown. Arrange young's modulus of materials in increasing order.



(1) $C < B < A < D$ (2) $D < C < B < A$

(3) $B < C < A < D$ (4) $A < B < C < D$

35. A string is wrapped around the rim of a wheel of moment of inertia 0.40 kg m^2 and radius 10 cm. The wheel is free to rotate about its axis. Initially the wheel is at rest. The string is now pulled by a constant force F. The angular velocity of the wheel after 10 s is 100 rad/s, the value of F is:

(1) 50 N (2) 40 N

(3) 120 N (4) 150 N

36. If the angular speed of earth becomes twice of its present value, then the weight of the person standing on north pole will be: (consider rotational effect of earth only)

(1) halved (2) doubled

(3) remain same (4) quadrupled

37. Consider the following statements:

(a) In straight-line motion with constant velocity, the angular momentum of a particle about any point in space remains constant.

(b) A couple (two equal and opposite forces, separated by some distance) always produces the same torque about any point in the plane of the forces.

(c) Angular momentum of an isolated system is conserved only when net external torque on the system is zero.

(d) In the perpendicular axis theorem, the axis must pass through the centre of mass of the body, otherwise the theorem cannot be applied.

Which of the above statements are **correct**?

(1) (a), (b) and (c) only

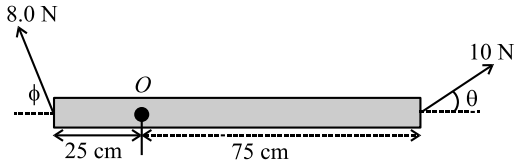
(2) (a) and (c) only

(3) (b), (c) and (d) only

(4) (a), (c) and (d) only

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38. The image shows a massless rod pivoted at the point *O*. A force of 8.0 N is applied at the left end, and a force of 10 N is applied at the right end, at the angles shown. If the rod is in rotational equilibrium, what is the relationship between the angles φ and θ?



(1) $\sin(\varphi) = 3.75 \sin(\theta)$

(2) $\sin(\theta) = 3.75 \sin(\varphi)$

(3) $\cos(\varphi) = 3.0 \sin(\theta)$

(4) $\sin(\varphi) = 3.0 \sin(\theta)$

39. A wire 8m in length suspended vertically stretches by 20 mm when mass of 3 kg is attached to the lower end. The elastic potential energy gain by the wire is: (take $g = 10\ m/s^2$);

(1) 0.3 J (2) 3 J

(3) 30 J (4) 300 J

40. Choose the **correct** statement/s.

Statement I- Gravitational force is independent of the medium, temperature and pressure.

Statement II- Gravitational force is conservative force.

Statement III- Gravitational force is a weak force. This force is significant mainly in large size heavy bodies.

Statement IV- Gravitational force may be repulsive in some cases.

(1) I, II and IV (2) I, II and III

(3) I, II, III and IV (4) II, III and IV

41. Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Bulk modulus of incompressible liquid is zero.

Reason R: The volume of incompressible liquid does not change by applying force.

In the light of the above statements, choose the **correct** answer from the options given below:

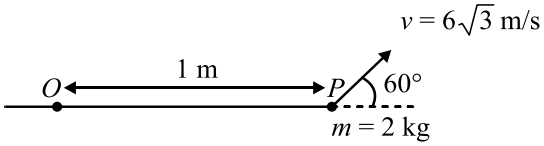
(1) A is true but R is false.

(2) A is false but R is true.

(3) Both A and R are true and R is the correct explanation of A.

(4) Both A and R are true but R is NOT the correct explanation of A.

42. The magnitude of angular momentum of a particle of mass 2 kg about point *O* at the instant as shown in the figure, is:



(1) $6\ kgm^2s^{-1}$

(2) $9\ kgm^2s^{-1}$

(3) $18\ kgm^2s^{-1}$

(4) $9\sqrt{3}\ kgm^2s^{-1}$

43. If the relative change in each edge of cube is 0.01, the volumetric strain is:

(1) 0.01

(2) 0.02

(3) 0.04

(4) 0.03

44. Two uniform identical discs rotating clockwise with angular velocities ω and 2ω separately about their axis of rotation passing through their centres are brought in contact. If the final angular velocity of the combined system is ω_f, then $\frac{\omega_f}{\omega}$ is:

(1) 1.5

(2) 2

(3) 0.5

(4) 4

45. Mass of thin uniform circular ring is *M* and its radius is *R*. Find the magnitude of maximum gravitational field on its axis;

(1) $\frac{2}{3\sqrt{3}}\frac{GM}{R^2}$

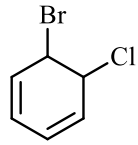
(2) $\frac{GM}{R^2}$

(3) $\frac{2GM}{R^2}$

(4) $\frac{3}{2\sqrt{2}}\frac{GM}{R^2}$

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46. Which of the following is **not** a state function?
(1) Temperature
(2) Enthalpy
(3) Work
(4) Internal energy.

47. The IUPAC name of the  is:

(1) 1-Bromo-2-chlorocyclohexa-3,5-diene
(2) 5-Bromo-6-chlorocyclohexa-1,3-diene
(3) 6-Bromo-5-chlorocyclohexa-1,3-diene
(4) 1-Bromo-6-chlorocyclohexa-2,4-diene

48. Work done during reversible isothermal expansion of one mole of ideal gas from 20 atm to 2 atm at 300 K is:
(1) –4.12 kJ
(2) –8.64 kJ
(3) –1.37 kJ
(4) –5.74 kJ

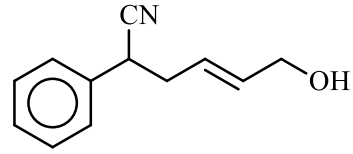
49. Match **List-I** with **List-II**.

List-I (Class of Compounds)		**List-II (IUPAC group prefix)**	
(A)	Esters	(I)	Carboxy
(B)	Carboxylic acids	(II)	Alkoxy
(C)	Ethers	(III)	Formyl
(D)	Aldehydes	(IV)	Alkoxycarbonyl

Choose the **correct** answer from the options given below:
(1) A-III; B-IV; C-II; D-I
(2) A-III; B-I; C-IV; D-II
(3) A-II; B-IV; C-I; D-III
(4) A-IV; B-I; C-II; D-III

50. An ideal gas expands against a constant external pressure of 1 atmosphere from 10 L to 40 L and absorbs 20 kJ of heat from surrounding. The change in internal energy of the system is:
(Given: 1 L-atm = 101.3 J)
(1) 23.01 kJ
(2) 16.96 kJ
(3) 19.24 kJ
(4) 11.71 kJ

51. IUPAC name of given compound is:



(1) 5-cyano-5-phenylpent-2-en-1-ol
(2) 6-cyano-5-phenylhex-2-en-1-ol
(3) 6-Hydroxy-2-phenylhex-4-en-1-nitrile
(4) 5-Hydroxy-1-phenylpent-3-en-1-nitrile

52. Consider the following processes:

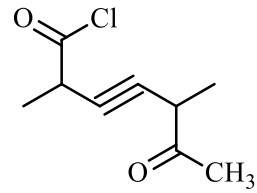
Process	$\Delta H(kJ\ mol^{-1})$	
$A \rightarrow 2B$	+80	… (1)
$B \rightarrow C + \frac{D}{2}$	–60	… (2)

For, $2C + D \rightarrow A$, ΔH will be:
(1) –70 kJ mol^{-1}
(2) +40 kJ mol^{-1}
(3) –40 kJ mol^{-1}
(4) +70 kJ mol^{-1}

53. $C_6H_5OCH_3$ is known as:
(1) Benzophenone
(2) Acetophenone
(3) Mesityl oxide
(4) Anisole

54. For which among the following species, standard molar enthalpy of formation at 298 K is **not** zero?
(1) $Br_2(g)$
(2) C(graphite)
(3) S(rhombic)
(4) $N_2(g)$

55. IUPAC name for the following compound is:



(1) 6-Chlorocarbonyl-3-methylhept-4-yn-2-one
(2) 7-Chlorocarbonyl-3,6-dimethylsept-4-yn-2-one
(3) 2,5-dimethyl-6-oxohept-3-yn-1-oyl chloride
(4) 5-Acetyl-2-methylhex-3-yn-1-oyl chloride

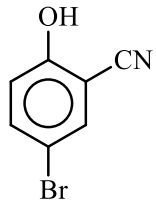
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56. Extensive property among the following is:
(1) density
(2) entropy
(3) pressure
(4) refractive index

57. The **correct** decreasing order of priority of functional groups in naming an organic compound as per IUPAC system of nomenclature is:
(1) $-CONH_2 > -\overset{O}{\overset{\|}{C}}- > -CHO$
(2) $-CHO > -\overset{O}{\overset{\|}{C}}- > -CONH_2$
(3) $-CHO > -CONH_2 > -\overset{O}{\overset{\|}{C}}-$
(4) $-CONH_2 > -CHO > -\overset{O}{\overset{\|}{C}}-$

58. Enthalpy of which process represents enthalpy of atomization?
(1) $CO_2(g) \rightarrow C(s) + O_2(g)$
(2) $CH_4(g) \rightarrow C(g) + 4H(g)$
(3) $C(s) + 2H_2(g) + \frac{1}{2}O_2(g) \rightarrow CH_3OH(\ell)$
(4) $2NH_3(g) \rightarrow N_2(g) + 3H_2(g)$

59. The IUPAC name of the following compound is:

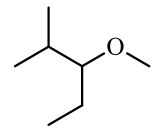
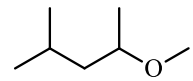
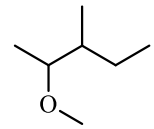
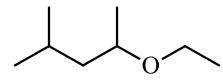


(1) 4-Bromo-2-cyanophenol
(2) 3-Bromo-6-hydroxybenzenecarbonitrile
(3) 3-Bromo-6-hydroxybenzonitrile
(4) 5-Bromo-2-hydroxybenzonitrile.

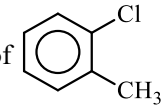
60. For a reaction at 298K, 2A + B → C, ΔH = 200 kJ mol^{-1} and ΔS = 0.4 kJ mol^{-1} K^{-1}. The minimum temperature above which the reaction will be spontaneous is:
(1) 327 °C
(2) 227 °C
(3) 27 °C
(4) 127 °C

61. IUPAC name of vinylacetylene is:
(1) Buta-1-ene-3-yne
(2) But-1-en-3-yne
(3) But-3-en-1-yne
(4) Buta-3-ene-1-yne

62. The **correct** option for a thermodynamic process in which a liquid kept inside a thermally insulated closed vessel at 25°C is mechanically stirred from outside is:
(1) $\Delta U < 0$ and $w > 0$
(2) $\Delta U < 0$ and $w < 0$
(3) $\Delta U > 0$ and $w > 0$
(4) $\Delta U > 0$ and $w < 0$

63. The **correct** structure of 2-methoxy-4-methylpentane is:
(1) 
(2) 
(3) 
(4) 

64. For a gaseous reaction,
$A(g) + 3B(g) \rightarrow 2C(g) + 2D(\ell)$:
ΔU is 15 kcal mol^{-1} at 27°C. The value of ΔH for the reaction will be:
(Given: R = 2 cal K^{-1} mol^{-1})
(1) 13.8 kcal mol^{-1}
(2) 16.2 kcal mol^{-1}
(3) Zero
(4) 18.5 kcal mol^{-1}

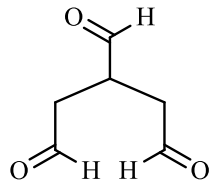
65. IUPAC name of  are:

(A) 1-chloro-2-methylbenzene
(B) 2-chlorotoluene
(C) 2-methylchlorobenzene
(1) A and B only (2) B and C only
(3) A and C only (4) A, B and C

[Contd...

66. Given below are two statements: one is labelled as Assertion A and other is labelled as Reason R:
Assertion A: No work is done during free expansion of an ideal gas during an irreversible process.
Reason R: Expansion of a gas in vacuum ($P_{ex} = 0$) is called free expansion.
In the light of above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true and R is NOT the correct explanation of A.

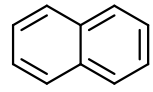
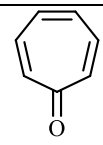
67. IUPAC name of given compound is:



(1) 3-Formylpentane-1,5-dial
(2) 2-Formylpropane-1,3-dial
(3) Propane-1,2,3-tricarbaldehyde
(4) Propane-1,2,3-trial

68. The $\Delta_r H$ of the given reaction at 298 K will be:
$C_2H_4(g) + H_2(g) \rightarrow C_2H_6(g)$
(Given: Bond energy of C–H, C–C, C=C and H–H bonds are 414, 347, 615 and 435 kJ mol^{-1} respectively)
(1) –150 kJ mol^{-1} (2) –100 kJ mol^{-1}
(3) –125 kJ mol^{-1} (4) –75 kJ mol^{-1}

69. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Tetrahydrofuran	(I)	
(B)	Tropone	(II)	
(C)	Thiophene	(III)	
(D)	Naphthalene	(IV)	

Choose the **correct** answer from the options given below:
(1) A-II; B-IV; C-I; D-III
(2) A-III; B-I; C-IV; D-II
(3) A-IV; B-I; C-II; D-III
(4) A-III; B-IV; C-II; D-I

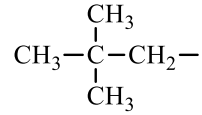
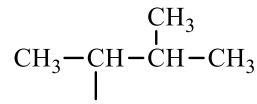
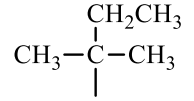
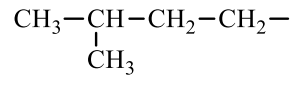
70. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Isothermal process	(I)	q = 0
(B)	Adiabatic process	(II)	w = 0
(C)	Isobaric process	(III)	$\Delta U = 0$
(D)	Isochoric process	(IV)	$\Delta P = 0$

Choose the **correct** answer from the options given below:
(1) A-III; B-I; C-IV; D-II
(2) A-II; B-IV; C-I; D-III
(3) A-IV; B-I; C-II; D-III
(4) A-III; B-IV; C-II; D-I

71. Identify the pair of compounds which are **not** member of same homologous series.
(1) CH_3CHO and CH_3COCH_3
(2) $CH{\equiv}CH$ and $CH_3C{\equiv}CH$
(3) CH_3OH and $CH_3CH_2CH_2OH$
(4) CH_4 and CH_3CH_3

72. Identify the process for which ΔS is positive.
(1) $2NO(g) + O_2(g) \rightarrow 2NO_2(g)$
(2) $2NaHCO_3(s) \rightarrow Na_2CO_3(s) + CO_2(g) + H_2O(g)$
(3) $2Cl(g) \rightarrow Cl_2(g)$
(4) $2HI(g) \rightarrow H_2(g) + I_2(g)$

73. Neo-pentyl group among the following is:
(1) 
(2) 
(3) 
(4) 

[Contd...

74. Given below are two statements:

Statement I: The heat content of the products is more than that of the reactants in an endothermic reaction.

Statement II: The entropy of any pure crystalline substance approaches zero as the temperature approaches 0°C.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

(1) Both Statement I and Statement II are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Both Statement I and Statement II are incorrect.

75. IUPAC name of ethylidene chloride is:

(1) 1,2-Dichloroethane
(2) 1,1-Dichloroethane
(3) 1-Chloroethene
(4) 1,2-Dichloroethene

76. For which of the following reactions, $\Delta_r H = \Delta_f H$?

(1) $H_2O(\ell) + \frac{1}{2}O_2(g) \rightarrow H_2O_2(\ell)$

(2) $2C(s) + O_2(g) \rightarrow 2CO(g)$

(3) $2B(s) + \frac{3}{2}O_2(g) \rightarrow B_2O_3(s)$

(4) $CO(g) + \frac{1}{2}O_2(g) \rightarrow CO_2(g)$

77. Match **List-I** with **List-II**.

List-I (Compound)		List-II (IUPAC name)	
(A)	Catechol	(I)	Benzene-1,3-diol
(B)	Resorcinol	(II)	Benzene-1,4-diol
(C)	Hydroquinone	(III)	4-methylphenol
(D)	p-cresol	(IV)	Benzene-1,2-diol

Choose the **correct** answer from the options given below:

(1) A-III; B-IV; C-II; D-I
(2) A-IV; B-I; C-II; D-III
(3) A-III; B-I; C-IV; D-II
(4) A-II; B-IV; C-I; D-III

78. Two mole of a gas is heated at constant pressure to raise its temperature by 10°C. The work done in joules is:

(1) –83.14 J
(2) –124.71 J
(3) –62.35 J
(4) –166.28 J

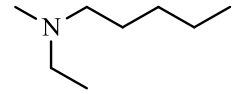
79. Hexanedioic acid is known as:

(1) Malonic acid
(2) Glutaric acid
(3) Adipic acid
(4) Succinic acid

80. The standard enthalpy of formation of $C_4H_{10}(g)$, $CO_2(g)$ and $H_2O(\ell)$ are x, y and z kJ mol^{-1} respectively. The standard enthalpy of combustion of butane (in kJ mol^{-1}) will be:

(1) 5y – 4z – x
(2) 2y + 3z – 2x
(3) x – 2y + 3z
(4) 4y + 5z – x

81. IUPAC name of given compound is:



(1) N-ethyl-N-methylpentan-1-amine
(2) N,N-ethylmethylpentan-1-amine
(3) Ethylmethylpentanamine
(4) Pentaneethylmethylamine

82. Given below are two statements:

Statement I: It is impossible to determine lattice enthalpies of ionic compounds directly by experiment.

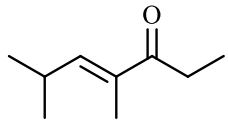
Statement II: For chemical reactions, heat absorbed at constant volume, is measured in a bomb calorimeter.

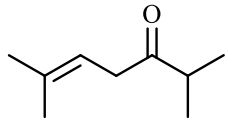
In the light of the above statements, choose the *most appropriate* answer from the options given below:

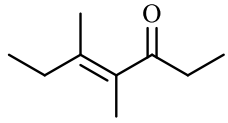
(1) Both Statement I and Statement II are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Both Statement I and Statement II are incorrect.

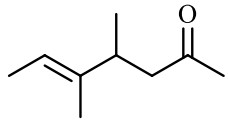
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83. Structure of 4,6-dimethylhept-4-en-3-one is:

(1) 

(2) 

(3) 

(4) 

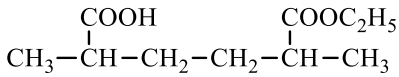
84. Match **List-I** with **List-II**.

List-I (Conditions)		List-II (Gibb's energy change)	
(A)	$\Delta H < 0$ and $\Delta S < 0$	(I)	$\Delta G > 0$ at all temperatures
(B)	$\Delta H > 0$ and $\Delta S > 0$	(II)	$\Delta G < 0$ at all temperature
(C)	$\Delta H < 0$ and $\Delta S > 0$	(III)	$\Delta G > 0$ at high temperature
(D)	$\Delta H > 0$ and $\Delta S < 0$	(IV)	$\Delta G > 0$ at low temperatures

Choose the **correct** answer from the options given below:

(1) A-IV; B-I; C-II; D-III
(2) A-III; B-IV; C-II; D-I
(3) A-III; B-I; C-IV; D-II
(4) A-II; B-IV; C-I; D-III

85. IUPAC name of given compound is:

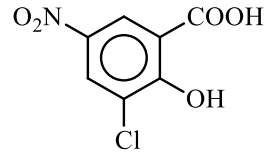


(1) 5-Ethoxycarbonyl-2-methylhexanoic acid
(2) 2-Carboxy-5-ethoxycarbonylhexane
(3) Ethyl-5-carboxy-2-methylhexanoate
(4) Ethyl-6-carboxy-2,5-dimethylhexanoate

86. For an irreversible Isothermal process:

(1) $\Delta S_{total} \neq 0$ and $\Delta U = 0$
(2) $\Delta S_{total} = 0$ and $\Delta U = 0$
(3) $\Delta S_{total} \neq 0$ and $\Delta U \neq 0$
(4) $\Delta S_{total} = 0$ and $\Delta U \neq 0$

87. IUPAC name of given compound is:

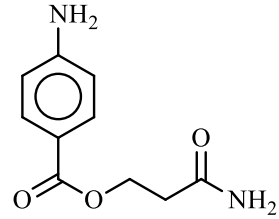


(1) 5-Nitro-3-chloro-2-hydroxybenzoic acid
(2) 3-Chloro-2-hydroxy-5-nitrobenzoic acid
(3) 3-Nitro-4-chloro-1-hydroxybenzoic acid
(4) 2-Hydroxy-3-chloro-5-nitrobenzoic acid

88. When 10 mole of an ideal gas is allowed to expand reversibly and adiabatically from a temperature of 27°C, the work done is 15 kJ mol^{-1}. The final temperature of the gas will be: (Given: C_v = 25 J mol^{-1} K^{-1})

(1) 200 K
(2) 240 K
(3) 260 K
(4) 220 K

89. The number of functional group present in the following compound is:



(1) 2
(2) 3
(3) 4
(4) 5

90. Given below are two statements: one is labelled as Assertion A and other is labelled as Reason R:
Assertion A: The solubility of most salts in water increases with rise of temperature.
Reason R: For most of ionic compounds, $\Delta_{sol}H°$ is positive and the dissolution process is endothermic.
In the light of above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true and R is NOT the correct explanation of A.

91. The **correctly** written scientific name of wheat is;
(1) triticum aestivum
(2) *Triticum aestivum*
(3) Triticum aestivum
(4) *Triticum Aestivum*

92. Given below are two statements:
Statement I: Viroids have RNA of low molecular weight that is surrounded by a protein coat.
Statement II: Prions are infectious agents consisting of abnormally folded protein, and are similar in size to viruses.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

93. All the following are related with Carolus Linnaeus, except;
(1) a two-kingdom classification system of organisms.
(2) proposal of binomial nomenclature system for naming of organisms.
(3) being awarded with the "*triple crown* of biology".
(4) giving the title of his publication as "*Systema naturae*".

94. How many of the following statements are **incorrect**?
A. In the five kingdom classification, bacteria are included in Kingdom Monera.
B. Diatoms are the chief 'producers' in the oceans.
C. The walls of diatoms are indestructible due to the presence of silica.
D. Red tides are caused by the rapid multiplication of red dinoflagellates such as *Gonyaulax*.
E. Slime moulds are saprophytic protists.
Choose the **correct** option.
(1) Two (2) Three
(3) Four (4) Zero

95. Match **List-I** with **List-II.**

List-I (Organism)		List-II (Family)	
(A)	Human	(I)	Anacardiaceae
(B)	Housefly	(II)	Solanaceae
(C)	Mango	(III)	Hominidae
(D)	*Petunia*	(IV)	Muscidae

Choose the *most appropriate* answer from the options given below.
(1) A-III, B-IV, C-II, D-I
(2) A-III, B-IV, C-I, D-II
(3) A-III, B-I, C-IV, D-II
(4) A-IV, B-III, C-I, D-II

96. A brain-dead patient in coma;
(1) doesn't grow.
(2) doesn't show metabolism.
(3) doesn't have self-consciousness.
(4) has all organs functioning normally.

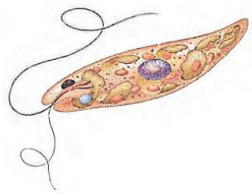
97. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A): The boundaries of Kingdom Protista are not well defined.
Reason (R): All protistans are primitive relative of animals.
In the light of the above statements, choose the *most appropriate* answer from the options given below.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

98. Consider the following statements.
A. As we go higher from species to kingdom, the number of common characteristics goes on decreasing.
B. Lower the taxa, more are the characteristics that the members within the taxon share.
C. Order is an assemblage of families which exhibit a few similar characters.
D. Genus comprises a group of related species which has less characters in common in comparison to species of other genera.
E. The family Felidae includes both cats and dogs.
The **correct** statements are;
(1) C, D and E
(2) A, B and C
(3) A, C and E
(4) B, C and D

99. The organisms that are able to produce biogas from dung of ruminants;
(1) are classified as true bacteria.
(2) have cell wall structure different from other bacteria.
(3) possess membrane-bound nucleus.
(4) can fix atmospheric nitrogen.

100. Identify the **incorrectly** matched pair.
(1) Genus *Panthera* - Includes lion and tiger
(2) Genus *Solanum* - Includes brinjal and potato
(3) Order Diptera - Includes housefly and gibbon
(4) Kingdom Animalia - Includes monkey and dog

101. Choose the correct statement about the organism shown below:



(1) It lacks cell membrane.
(2) It is parasitic in nutrition.
(3) It is a unicellular eukaryote.
(4) It produces spores.

102. Go through the taxonomical hierarchy given below.

Kingdom
↓
A
↓
Class
↓
B
↓
C
↓
Genus
↓
D

Choose the **correct** statement about it.
(1) For animals 'A' is phylum and for plants it is 'Division'.
(2) 'B' includes several classes.
(3) The category 'C' of *Datura* is included in order Poales.
(4) *Mangifera* and *Felis* belong to same 'D'.

103. Match **List-I** with **List-II.**

List-I		List-II	
(A)	Amoeboid protozoans	(I)	One of the parasitic forms causes sleeping sickness
(B)	Flagellated protozoans	(II)	Have a gullet that opens to the outside of the cell surface
(C)	Ciliated protozoans	(III)	One member is the malarial parasite
(D)	Sporozoans	(IV)	Marine forms have silica shells on their surface

Choose the *most appropriate* answer from the options given below.
(1) A-IV, B-II, C-I, D-III
(2) A-I, B-IV, C-II, D-III
(3) A-IV, B-I, C-II, D-III
(4) A-IV, B-I, C-III, D-II

104. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A): During sexual reproduction in ascomycetes and basidiomycetes, an intervening dikaryotic stage occurs.
Reason (R): During sexual reproduction, in these fungi, plasmogamy is immediately followed by karyogamy, leading to diploid cells.
In the light of the above statements, choose the *most appropriate* answer from the options given below.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

105. Read the following statements w.r.t fungi.
A. The cell walls of fungi are composed of chitin and polysaccharides.
B. Yeast, a unicellular fungus, is used to make bread and beer.
C. The network of hyphae is known as mycelium.
D. Hyphae that are continuous tubes filled with multinucleated cytoplasm are called coenocytic hyphae.
E. Most fungi are autotrophic and absorb inorganic matter.
The **incorrect** statement(s) is /are;
(1) A, B, C
(2) E only
(3) B, C, D
(4) C and E

106. The earliest classifications were based on:
(1) phylogenetic relationships.
(2) the 'uses' of various organisms.
(3) mode of nutrition.
(4) cell structure.

107. Bladderwort and *Cuscuta* are examples of:
(1) insectivorous and parasitic plants, respectively.
(2) parasitic and insectivorous plants, respectively.
(3) saprophytic plants.
(4) unicellular autotrophs.

[Contd...

108. Given below are two statements:

Statement I: Consciousness is a defining property of living organisms.

Statement II: Human being is the only organism who is aware of himself, i.e., has self-consciousness.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

109. In the five kingdom classification proposed by R.H. Whittaker, the tetanus pathogen and bread mould were, respectively, placed in kingdom;

(1) Fungi and Protista.
(2) Monera and Fungi.
(3) Protista and Monera.
(4) Plantae and Animalia.

110. The recognition that all present-day living organisms are related to each other and to all organisms that ever lived on this earth was a revelation that led to:

(1) anthropocentric view of biology.
(2) limited progress in biological knowledge.
(3) cultural movements for conservation of biodiversity.
(4) deification of inanimate matter.

111. Match **List-I** with **List-II.**

List-I		List-II	
(A)	*Mucor*	(I)	Protein is the infectious material
(B)	*Anabaena*	(II)	Septate and branched hyphae
(C)	*Penicillium*	(III)	Aseptate and coenocytic mycelium
(D)	Prions	(IV)	Possess heterocysts

Choose the *most appropriate* answer from the options given below.

(1) A-III, B-IV, C-II, D-I
(2) A-III, B-II, C-IV, D-I
(3) A-I, B-IV, C-II, D-III
(4) A-III, B-IV, C-I, D-II

112. Read the following statements about viruses.

A. Viruses are non-cellular organisms.
B. W.M. Stanley demonstrated that the extract of infected tobacco plants could cause infection in healthy plants.
C. No virus contains both RNA and DNA.
D. The protein coat called capsid is made of small subunits called capsomeres.
E. Bacteriophages that infect bacteria are usually single-stranded DNA viruses.

In the light of above statements choose the **correct** answer from the options given below:

(1) A, B, C are incorrect
(2) A, C, D are correct
(3) A, C, E are incorrect
(4) B, D, E are correct

113. All the following are true for pellicle, **except** that it;

(1) is present in *Euglena*.
(2) makes the body of euglenoids flexible.
(3) is surrounded by a cell wall.
(4) is a protein rich layer.

114. Select the **incorrect** pair of a disease and its causative agent group.

(1) Mad cow disease – Prion
(2) Cholera – Bacterium
(3) Mumps – Virus
(4) Citrus canker – Viroid

115. The first step in the sexual cycle of fungi is;

(1) karyogamy.
(2) fusion of protoplasms between two motile or non-motile gametes.
(3) fusion of parental nuclei.
(4) meiosis in zygote.

116. Which of the following is a **correct** set of asexual spores in fungi?

(1) Zygospore and conidia
(2) Conidia and zoospore
(3) Aplanospore and oospore
(4) Sporangiospore and basidiospore

117. Choose the **incorrect** statement.

(1) Classification is a single step process.
(2) Each taxonomical category is a part of overall taxonomic arrangement.
(3) Each taxonomical category is referred to as a unit of classification.
(4) Each taxonomical category represents a rank and is commonly termed as taxon.

118. Read the following statements.

A. Increase in mass is a defining characteristic of living organisms.

B. Reproduction is seen invariably in all living organisms.

C. Metabolism is observed in living organisms as well as non-living objects.

D. All living organisms can sense their surroundings or environment.

E. Sand-mounds grow by accumulation of material on the surface.

The number of **incorrect** statements is;

(1) five. (2) three.

(3) four. (4) zero.

119. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A): Systematics takes into account evolutionary relationships between organisms.

Reason (R): The word systematics is derived from the Latin word 'systema' which means systematic arrangement of organisms.

In the light of the above statements, choose the *most appropriate* answer from the options given below.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(3) (A) is true but (R) is false.

(4) (A) is false but (R) is true.

120. Morels and truffles are members of which class of Fungi?

(1) Phycomycetes (2) Deuteromycetes

(3) Basidiomycetes (4) Ascomycetes

121. Which of the following sets belong to the same taxonomic category?

(1) *Homo* and *sapiens*

(2) Mammalia and Insecta

(3) Angiospermae and Dicotyledonae

(4) Carnivora and Canidae

122. Given below are two statements:

Statement I: Whittaker's five-kingdom classification was based on cell structure, body organisation, mode of nutrition, reproduction, and phylogenetic relationships.

Statement II: The two-kingdom system of classification was found inadequate because it did not distinguish between eukaryotes and prokaryotes.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both Statement I and Statement II are correct.

(2) Both Statement I and Statement II are incorrect.

(3) Statement I is correct but Statement II is incorrect.

(4) Statement I is incorrect but Statement II is correct.

123. At which rank do the humans and dog differ?

(1) Phylum

(2) Class

(3) Order

(4) Kingdom

124. The organisms that can survive in extreme salty areas are;

(1) halophiles.

(2) thermoacidophiles.

(3) cyanobacteria.

(4) sporozoans.

125. Match **List-I** with **List-II.**

List-I		List-II	
(A)	Mycoplasma	(I)	Lack cell wall
(B)	Desmids	(II)	Didn't find a place in Whittaker's classification
(C)	Common mushroom	(III)	Golden algae
(D)	Lichens	(IV)	Edible fungus

Choose the *most appropriate* answer from the options given below.

(1) A-III, B-I, C-IV, D-II

(2) A-I, B-III, C-IV, D-II

(3) A-III, B-IV, C-I, D-II

(4) A-III, B-I, C-II, D-IV

126. Among the following, the lowest taxonomical category is;

(1) Monocotyledonae.

(2) Arthropoda.

(3) Muscidae.

(4) Animalia.

127. In a lichen, the phycobiont is;

(1) a heterotrophic organism.

(2) able to synthesise food.

(3) responsible to provide shelter to its partner.

(4) an acellular organism.

[Contd...

128. Read the given statements about tobacco mosaic virus.

A. It possesses DNA as genetic material.
B. It lacks capsid.
C. It consists of head, collar, sheath and tail fibres.
D. It is inert outside its host cell.
E. It can pass through bacteria-proof filters.

Choose the **correct** option.

(1) A, B and C are correct
(2) B, C and D are incorrect
(3) A, D and E are incorrect
(4) Only D and E are correct

129. Choose the **incorrect** statement.

(1) In yeast, budding is observed.
(2) Hydra can reproduce by budding.
(3) *Planaria* does not reproduce.
(4) In *Amoeba*, reproduction is synonymous with growth.

130. In the organism shown below:



(1) sexual spores are called ascospores.
(2) sex organs are not present.
(3) sexual reproduction is not seen.
(4) conidia are not produced.

131. Match **List-I** with **List-II.**

List-I		List-II	
(A)	Number of components in a scientific name	(I)	4
(B)	Biodiversity estimated on earth	(II)	7
(C)	Number of broad categories in taxonomical hierarchy	(III)	1.7-1.8 million
(D)	Number of classes in kingdom Fungi	(IV)	2

Choose the *most appropriate* answer from the options given below.

(1) A-IV, B-III, C-II, D-I
(2) A-IV, B-II, C-III, D-I
(3) A-I, B-III, C-II, D-IV
(4) A-III, B-IV, C-I, D-II

132. How many kingdoms in Whittaker's system of classification contain eukaryotes?

(1) One only
(2) Two only
(3) Three only
(4) Four only

133. The *'Contagium vivum fluidum'* (infectious living fluid) was a term coined by:

(1) Dmitri Ivanowsky
(2) W.M. Stanley
(3) T.O. Diener
(4) M.W. Beijerinck

134. Consider the following statements:

A. Aristotle was the earliest to attempt a more scientific basis for classification.
B. Aristotle classified plants into trees, shrubs, and herbs.
C. Whittaker classified viroids as monerans.
D. The two-kingdom classification system placed fungi in the plant kingdom.
E. Linnaeus divided animals into those with red blood and those without it.

The **correct** statements are;

(1) C, D and E
(2) A, B and D
(3) B, C and D
(4) A, C and E

135. Lichens are considered good indicators of pollution because they:

(1) grow very fast in polluted areas.
(2) are symbiotic associations.
(3) do not grow in polluted areas.
(4) cause blooms in polluted water bodies.

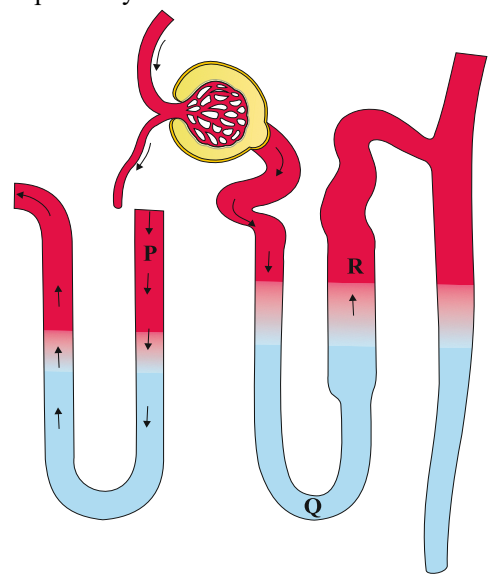
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136. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Smooth muscle fibers are described as non-striated because they lack the distinct, alternating light and dark bands seen in skeletal and cardiac muscle.
Reason R: The absence of striations in smooth muscle means that they completely lack the contractile proteins actin and myosin.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is not the correct explanation of A.

137. Identify the **correct** sequence of True (T) or False (F) statements:
I. Liver plays a significant role in excretion by converting highly toxic ammonia into less toxic urea.
II. In the process of dialysis, the dialysing fluid is isotonic to blood plasma except for the absence of nitrogenous wastes.
III. Glomerular filtration is highly selective, preventing the filtration of even minute molecules.
IV. ANF (Atrial Natriuretic Factor) causes vasoconstriction, leading to an increase in blood pressure and GFR.
(1) F, T, T, T (2) T, T, F, F
(3) T, F, F, T (4) F, T, F, T

138. Given below are two statements.
Statement I: The myofibril is a bundle of actin and myosin filaments, while the muscle fiber (muscle cell) is a bundle of myofibrils.
Statement II: Red muscle fibers are rich in myoglobin and mitochondria, primarily performing aerobic respiration, whereas white muscle fibers rely mainly on anaerobic respiration.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

139. What is the **correct** representation of the osmolarity (in mOsmol/L) of the filtrate/fluid at the P, Q and R respectively?



(1) 300,1200,300
(2) 300,1200,100
(3) 1200,300,100
(4) 100,1200,300

140. Match **List-I** with **List-II**.

List-I (Bone)		List-II (Body Part)	
(A)	Scapula	(I)	Forearm
(B)	Tarsals	(II)	Shoulder Girdle
(C)	Tibia	(III)	Ankle/Foot
(D)	Radius	(IV)	Lower Leg

Choose the **correct** answer from the options given below:
(1) A-II, B-IV, C-III, D-I
(2) A-II, B-III, C-IV, D-I
(3) A-I, B-II, C-IV, D-III
(4) A-III, B-IV, C-I, D-II

141. A new drug is introduced that selectively and completely inhibits the active transport of NaCl in the ascending limb of the loop of Henle. What would be the immediate and most significant consequence on the human kidney?
(1) A significant increase in GFR due to reduced efferent arteriole resistance.
(2) The filtrate in the DCT would become hypertonic to the plasma.
(3) The maximum medullary osmotic gradient would be significantly reduced from 1200 mOsmol/L.
(4) ADH would cease to have any effect on water reabsorption.

[Contd...

142. How many of the following statements are **correct**?
I. Cardiac muscles are striated, involuntary, and possess intercalated discs at their fusion points.
II. Tetany is rapid spasms (wild contractions) in muscle due to low Ca^{++} in body fluid.
III. The A -band in a sarcomere comprises the entire length of the thick filaments and the overlapping portions of the thin filaments.
IV. Synovial joints are lubricated by a viscous fluid secreted by the articular cartilage in the lacunae.
V. Fibrous joints are characterised by the absence of movement between the articulating bones.
(1) Three (2) Four
(3) Five (4) Two

143. Which of the following is the **correct** number of bones in the human vertebral column?
(1) 7 Cervical, 10 Thoracic, 5 Lumbar, 5 Sacral (not fused), 4 Coccygeal (not fused)
(2) 7 Cervical, 12 Thoracic, 5 Lumbar, 1 (fused) Sacral, 1 (fused) Coccygeal
(3) 7 Cervical, 10 Thoracic, 5 Lumbar, 1 Sacral, 1 Coccygeal
(4) 7 Cervical, 12 Thoracic, 5 Lumbar, 4 Sacral, 4 Coccygeal

144. Match **List-I** with **List-II**.

List I		List II	
(A)	Excretory organ in *Amphioxus*	(I)	Bony Fishes
(B)	Human kidney	(II)	Flame cells
(C)	Ammonotelic	(III)	Crocodiles
(D)	Uricotelic	(IV)	10-12 cm in length

Choose the **correct** answer from the options given below:
(1) A-II, B-IV, C-I, D-III
(2) A-IV, B-II, C-III, D-I
(3) A-II, B-III, C-I, D-IV
(4) A-I, B-IV, C-II, D-III

145. Given below are two statements.
Statement I: The myosin head acts as an ATPase enzyme, hydrolysing ATP into ADP and inorganic phosphate Pi to generate the energy required for the rotation of myosin head.
Statement II: The rotation of myosin head occurs when the myosin head detaches from the actin filament.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement I is correct but Statement II is incorrect.
(2) Both Statement I and Statement II are correct.
(3) Statement I is incorrect but Statement II is correct.
(4) Both Statement I and Statement II are incorrect.

146. How many of the following statements are **correct**?
A. The kidney's counter-current mechanism is highly effective in conserving water and producing concentrated urine.
B. NaCl is reabsorbed by the Loop of Henle and urea is reabsorbed by the collecting duct into the medullary interstitium, both contributing to the high osmolarity.
C. The facultative reabsorption of water occurs primarily in the collecting duct and the distal convoluted tubule.
D. The excretion of nitrogenous waste in the form of urea in mammals is an adaptation to conserve water relative to ammonotelic organisms.
E. Urea is significantly less toxic than ammonia, requiring a smaller volume of water for its removal from the body.
(1) Three (2) Four
(3) Five (4) Two

147. Considering the role of Ca^{2+} in skeletal muscle contraction, which of the following sequences **best** describes the events following the binding of Ca^{2+} to its target protein?
(1) Ca^{2+} binds to Actin $\rightarrow$ Tropomyosin shifts $\rightarrow$ Myosin binding sites are exposed.
(2) Ca^{2+} binds to Troponin $\rightarrow$ Troponin complex changes conformation $\rightarrow$ Tropomyosin moves $\rightarrow$ Active sites on Actin are uncovered.
(3) Ca^{2+} binds to Tropomyosin $\rightarrow$ Myosin head detaches $\rightarrow$ ATP hydrolysis occurs.
(4) Ca^{2+} binds to the Z-line $\rightarrow$ Actin filaments shorten $\rightarrow$ Muscle relaxes.

148. Which of the following pairings of bones forming the pelvic girdle is **correct**?
(1) Ilium, Ischium, Pubis, and Femur
(2) Ilium, Acetabulum, Ischium, and Pubis
(3) Two Coxal Bones (each formed by the fusion of Ilium, Ischium, and Pubis)
(4) Ilium, Sacrum and Ischium

149. What substance is **not** typically found in sweat produced by sweat glands?
(1) NaCl
(2) Urea
(3) Lactic acid
(4) Cholesterol

[Contd...

150. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: The muscle may remain in contracted state due to unavailability of ATP.

Reason R: The bond between the myosin head and actin is broken by the binding of a fresh ATP molecule.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is not the correct explanation of A.

151. What role do sebaceous glands play in the skin?

(1) Secrete bile to aid digestion
(2) Produce sweat to cool the body
(3) Eliminate substances like sterols and hydrocarbons through sebum
(4) Remove CO_2 and water

152. What type of joint is present between the Humerus and the Ulna and what movement does it primarily allow?

(1) Pivot Joint; Rotation
(2) Hinge Joint; movement in single plane
(3) Ball and Socket Joint; movement in multiple directions
(4) Condyloid Joint; Biaxial movement

153. Analyze the following statements about bone structure and classification:

I. The number of bones in the human skull is 22 (Cranial + Facial).
II. The pectoral girdle and the pelvic girdle are part of the appendicular skeleton.
III. All synovial joints are biaxial, allowing movement in two perpendicular planes.
IV. Chondrocytes are bone-forming cells responsible for deposition of the bone matrix.

Which of the above statement is/are **correct**?

(1) I, II and IV only
(2) II, III and IV only
(3) I and II only
(4) II and IV only

154. Considering the detailed structural arrangement of the Glomerular Filtration Membrane (GFM), what is the **correct** sequence of layers or their components that the filtrate must pass through?

(1) Podocytes → Basement Membrane → Endothelium of Glomerular Capillary
(2) Endothelium of glomerular blood vessels→ Basement Membrane → Epithelium of Bowman's capsule
(3) Endothelium → Filtration Slits → Basement Membrane
(4) Podocytes → Basement Membrane → Afferent Arteriole

155. Mark the **correct** statement.

(1) Limb bones such as carpals serves weight bearing functions.
(2) Movement of particles and mucus in a specific direction is the function of squamous epithelium.
(3) In all connective tissues except bones, the cell secrete fibres.
(4) Cell junctions fuse the plasma membranes of cardiac muscle cells and make them stick together.

156. Consider the statements.

A. In diabetes mellitus, the urine contains glucose and ketone bodies.
B. In diabetes insipidus the volume of urine output increases due to efficiency of ADH.
C. If the collecting duct of nephron does not respond to ADH the dilution of urine occurs even if the level of ADH is normal.
D. Collecting Duct allows passage of small amounts of urea into the medullary interstitium to keep up the osmolarity.

Which of the following option represents the combination of **incorrect** statements?

(1) A and B only (2) B and C only
(3) B only (4) B, C and D only

157. Considering the fate of filtered substances in a healthy human nephron, which statement is physiologically **accurate**?

(1) Urea is reabsorbed primarily in the DCT and actively secreted into the PCT.
(2) Nearly all of the essential nutrients, and 70-80 per cent of water and no electrolytes are reabsorbed by PCT.
(3) Glucose is primarily reabsorbed by facilitated diffusion in the PCT and DCT.
(4) The reabsorption of bicarbonate ions (HCO_3^-), which is essential for maintaining blood pH, occurs in both the PCT and DCT.

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158. Given below are two statements.

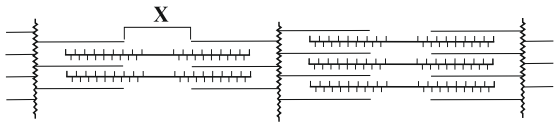
Statement I: In the early stages of the nephron, the glomerular filtrate is almost isotonic to blood plasma, despite the complete absence of plasma proteins in the glomerular filtrate.

Statement II: Although proteins are absent, the concentration of all filterable solutes (electrolytes, glucose, urea, etc.) in the glomerular filtrate is almost identical to their concentration in the plasma.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

159.



In the above shown diagram, a cross-section through the 'X' would reveal which arrangement of myofilaments?

(1) Only thick Myosin filaments.
(2) Only thin Actin filaments.
(3) Both thick and thin filaments in an overlapping arrangement.
(4) Only Actin filaments anchored by the Z-line protein.

160. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Micturition reflex i.e., the neural mechanism causing micturition, is a voluntary action.

Reason R: The stretch receptors on the walls of the urinary bladder send signals to the CNS when the bladder is filled. The voluntary control is achieved by the conscious relaxation or contraction of the external urethral sphincter.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is not the correct explanation of A.
(3) A is true but R is false.
(4) A is false but R is true.

161. If a section of a myofibril is labeled with Z, A, I, H, and M lines, what is the sequence of bands/zones that would be crossed if an injected molecule travelled from the centre of one sarcomere to the centre of an adjacent sarcomere?

(1) M→H→A→Z→I→Z→A
(2) M→H→A→I→Z→I→A
(3) M→H→A→I→Z→A
(4) M→H→A→I→Z→I→A→H→M

162. Given below are two statements.

Statement I: Urethra is the duct that carries urine from the renal pelvis to the urinary bladder.

Statement II: Distal convoluted tubule is primarily responsible for maintaining the pH and ionic balance of the blood by tubular secretion of H^+, K^+, and ammonia.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

163. Consider the following statements.

I. Cortical nephrons are more numerous than juxtamedullary nephrons.
II. Cortical nephrons have a very short loop of Henle which mostly extends only into the renal medulla.
III. The concentration of urea is high in the renal artery and lowest in the renal vein.
IV. The *vasa recta* is absent or highly reduced in cortical nephrons.
V. In a state of acidosis (low blood pH), the kidney's mechanism to restore pH involves increased secretion of H^+ and increased reabsorption of HCO_3^-.

Which of the above statement is/are **correct**?

(1) I, II and V only
(2) II, III and IV only
(3) II and IV only
(4) I, III, IV and V only

[Contd...

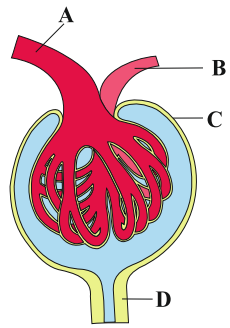
164. Match **List-I** with **List-II**.

List-I		List-II	
(A)	*Paramoecium*	(I)	Uses tentacles for capturing its prey
(B)	*Hydra*	(II)	Cilia helps in movement of food through cytopharynx
(C)	*Vasa recta*	(III)	Helps in excretion in insects
(D)	Malpighian Tubules	(IV)	Counter-current exchanger

Choose the **correct** answer from the options given below:

(1) A-I, B-II, C-IV, D-III
(2) A-II, B-I, C-IV, D-III
(3) A-II, B-I, C-III, D-IV
(4) A-I, B-II, C-III, D-IV

165. Difference between the diameter of which two labelled parts among the following are responsible for ultrafiltration.



(1) A and C (2) B and D
(3) A and D (4) A and B

166. A patient is diagnosed with severe renal failure, necessitating dialysis. The principle of dialysis (artificial kidney) is most closely based on which physiological process?

(1) Active transport and -pump action.
(2) Counter-current mechanism and hormonal regulation.
(3) Diffusion and Ultrafiltration across a semi-permeable membrane.
(4) Secondary active transport

167. Given below are two statements.

Statement I: Each organised skeletal muscle in our body is made of a number of muscle bundles or fascicles held together by a common collagenous connective tissue layer called fascia.

Statement II: Contractility is the ability of a muscle fiber that makes it able to rebound to its original length after contraction.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

168. A significant increase in glomerular blood flow/pressure to the juxtaglomerular apparatus causes the release of a peptide hormone from the atrial wall of the heart. Identify the **correct** function of this hormone.

(1) It stimulates the release of Aldosterone, increasing Na^+ reabsorption.
(2) It acts as a check on renin-angiotensin mechanism and caused vasodilation.
(3) It increases the permeability of the collecting duct to water, promoting water reabsorption.
(4) It constricts the afferent arteriole, decreasing GFR.

169. Which type of movement is employed by the macrophages and leucocytes in the blood?

(1) Muscular movement
(2) Ciliary movement
(3) Flagellar movement
(4) Amoeboid movement

170. Select the **incorrectly** matched pair regarding excretory structures in different organisms:

(1) Protonephridia - Rotifers
(2) Nephridia - Prawns
(3) Antennal Glands - Crustaceans
(4) Flame Cells – *Planaria*

171. Match the muscle disorders in **List-I** with their key characteristic in **List-II**.

List-I		List-II	
(A)	Myasthenia Gravis	(I)	Decreased level of oestrogen is a common cause
(B)	Muscular Dystrophy	(II)	Autoimmune disorder affecting neuromuscular junction
(C)	Osteoporosis	(III)	Progressive degeneration of skeletal muscle
(D)	Gout	(IV)	Accumulation of uric acid crystals in joints

Choose the **correct** answer from the options given below:

(1) A-I, B-II, C-III, D-IV
(2) A-II, B-III, C-I, D-IV
(3) A-III, B-I, C-II, D-IV
(4) A-I, B-III, C-II, D-IV

[Contd...

172. A pharmaceutical company develops a new drug that specifically inhibits the conversion of angiotensin I to angiotensin II. Which of the following effects is the **most** likely physiological consequence of administering this drug?
(1) Increased secretion of ADH leading to water retention and increased GFR.
(2) Increased conversion of Angiotensin II, leading to vasoconstriction and increased BP.
(3) Reduced production of Aldosterone, resulting in decreased Na^+ reabsorption and decreased BP.
(4) Reduced release of Renin from the JG cells, leading to decreased GFR.

173. Carefully read the following statements and identify how many of the following statements are **correct**:
A. The basic functional unit of a skeletal muscle fiber spans from one Z-line to the next Z-line.
B. Each vertebra has a central hollow portion through which the spinal cord passes.
C. First vertebra is the axis and it articulates with the occipital condyles.
D. All locomotion involves movement, but all movement is not locomotion.
E. Bones primarily are not composed of calcium.
(1) Only three statements are correct
(2) Only four statements are correct
(3) All five statements are correct
(4) Only two statements are correct

174. Which of the following is the most direct consequence of a failure of energy source needed for active transport in the cells of the Proximal Convoluted Tubule (PCT)?
(1) Increased glucose reabsorption leading to glycosuria.
(2) A reduction in the driving force for Na^+ reabsorption, impairing the active transport of glucose and amino acids.
(3) Excessive water reabsorption due to a sustained high osmotic gradient.
(4) Increased secretion of K^+ and H^+ into the tubular filtrate.

175. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Pivot Joint	(I)	Between Carpals
(B)	Gliding Joint	(II)	Between Atlas and Axis
(C)	Saddle Joint	(III)	Between Humerus and Pectoral Girdle
(D)	Ball and Socket Joint	(IV)	Between Carpal and Metacarpal of Thumb

Choose the **correct** answer from the options given below:
(1) A-II, B-I, C-IV, D-III
(2) A-I, B-II, C-IV, D-III
(3) A-II, B-IV, C-I, D-III
(4) A-III, B-II, C-I, D-IV

176. Which of the following options is/are an example of synovial joints?
A. Joint between the humerus and pectoral girdle.
B. Knee joint.
C. Joint between the adjacent vertebrae.
D. Joint between atlas and axis.
(1) A, B and D
(2) B and C
(3) C only
(4) D only

177. How many of the following statements are **correct**?
A. Juxtaglomerular (JG) cells are modified smooth muscle cells that act as baroreceptors, releasing Renin when afferent arteriole BP drops.
B. The kidney is the primary organ for maintaining electrolyte balance and acid-base balance.
C. Patella is an elbow cap.
D. Towards the centre of the inner concave surface of the kidney is a notch called hilum is present through which ureter, blood vessels and nerves enter.
E. The sarcolemma is the plasma membrane of a muscle fibre.
(1) Three (2) Four
(3) Five (4) Two

178. Given below are two statements.
Statement I: A motor neuron along with the muscle fibres connected to it constitute a motor unit.
Statement II: In the sliding filament theory of muscle contraction, the A -band width remains constant while the I-band and H-zone shorten.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Statement I is incorrect but Statement II is correct.
(3) Statement I is correct but Statement II is incorrect.
(4) Both Statement I and Statement II are incorrect.

[Contd...

179. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: The release of Angiotensin II causes a potent vasoconstriction which helps in restoring Glomerular Filtration Rate (GFR).

Reason R: Angiotensin II directly stimulates the juxtaglomerular cells (JGC) to release Aldosterone, which acts on the PCT to increase Na^+ reabsorption.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is not the correct explanation of A.

180. Which of the following bone is **not** attached with any other bone directly?

(1) Femur
(2) Humerus
(3) Hyoid bone
(4) Ulna

■■■

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